IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of Methodologies

- Data Collection
- Data Wrangling
- Exploratory Data Analysis with Data Visualization
- Exploratory Data Analysis Using SQL
- Interactive Mapping with Folium
- Predictive Analysis (Classification)

Summary of Results

- Exploratory Data Analysis Results
- Predictive Analysis Results

Introduction

Project Background & Context

SpaceX is the most successful company of the commercial space age, making space travel significantly more affordable. The company advertises Falcon 9 rocket launches on its website at a cost of $62 million, whereas other providers charge upward of $165 million per launch. Much of these savings stems from SpaceX's ability to reuse the rocket's first stage. Therefore, if we can determine whether the first stage will land, we can estimate the cost of a launch. Using public data and machine learning models, this project aims to predict whether SpaceX will successfully land and reuse the first stage.

Research Questions

1. How do variables such as payload mass, launch site, number of flights, and orbit type affect the success of a first-stage landing?

2. Has the rate of successful landings increased over time?

3. Which binary classification algorithm yields the best predictive performance in this case?

Section 1

Methodology



Methodology

Data Collection & Methodology

Data Collection

- Retrieving data via the SpaceX REST API
- Web-scraping data from Wikipedia

Data Wrangling

- Filtering the dataset
- Handling missing values
- Applying one-hot encoding to prepare data for classification

Exploratory Data Analysis (EDA)

- Conducting EDA using data visualization and SQL queries

Interactive Visual Analytics

- Developing interactive visualizations and dashboards using Folium and Plotly Dash

Predictive Analysis

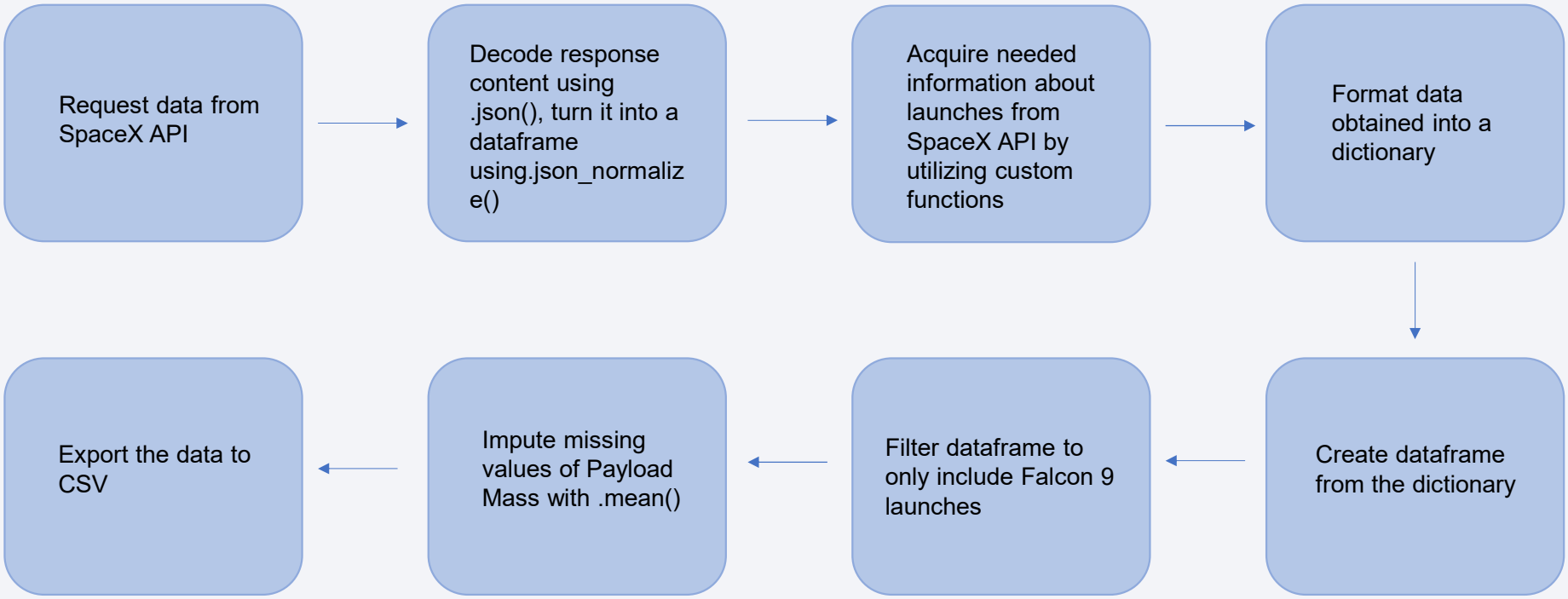
- Building, tuning, and evaluating classification models for optimal performance

Data Collection

- The data collection process combined requests to the SpaceX REST API (which is now inaccessible) with web scraping from a Wikipedia page containing SpaceX launch records. Combining these sources provided a comprehensive dataset for a more detailed analysis.
- The columns acquired from the SpaceX REST API included: FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, and Latitude.
- The columns acquired from the Wikipedia table included: Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, and Time.

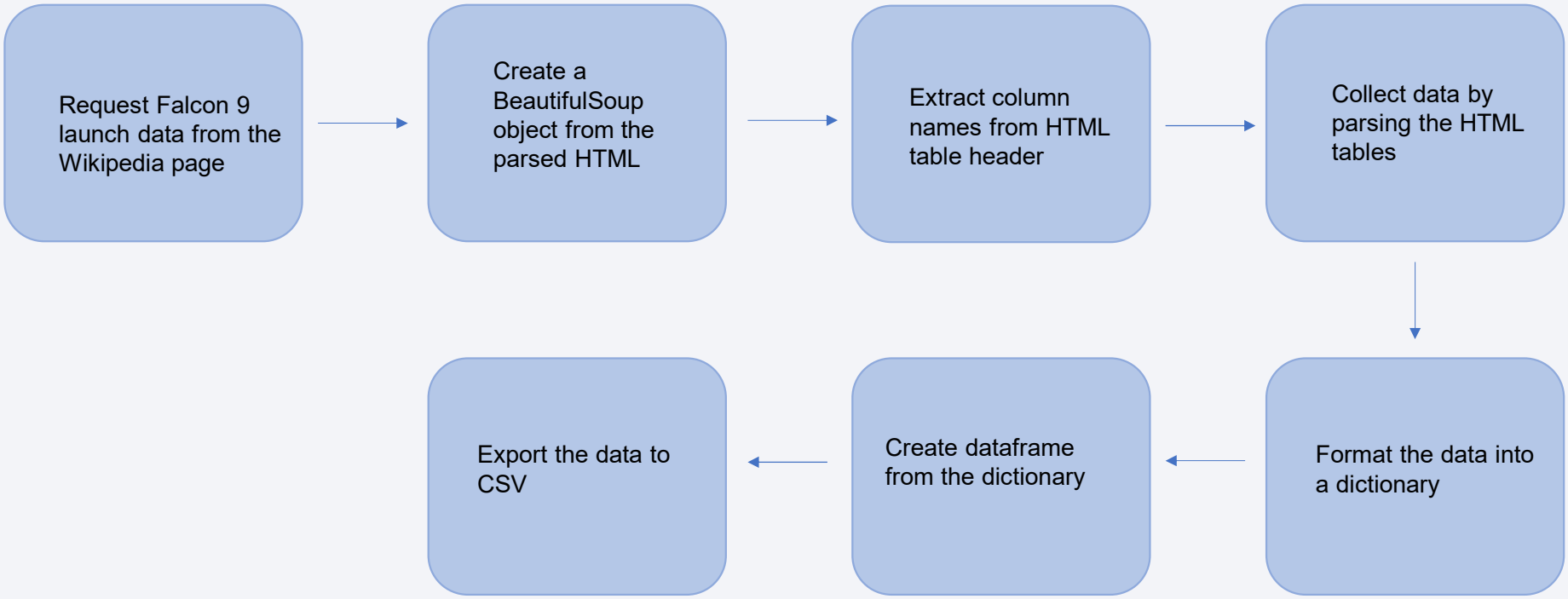
Data Collection – SpaceX API

GitHub Link



Data Collection – Web Scraping

GitHub Link



Data Wrangling

GitHub Link

- Performed EDA and determined training labels.
- Calculated the number of occurrences at each launch site.
- Calculated the number of occurrences of each orbit.
- Calculated the number of occurrences of each mission.
- Created a landing outcome label from the existing outcome column.
- Exported the data to CSV.

EDA with Data Visualization

GitHub Link

The following charts were plotted

- Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload Mass vs. Launch Site, Orbit Type vs. Success Rate, Flight Number vs. Orbit Type, Payload Mass vs. Orbit Type, Success Rate Yearly Trend.
- Scatter plots demonstrate relationships between variables. Relationships between variables can be leveraged in machine learning models. Bar charts compare categorical data, and the goal was to show relationships between specific categories against a measured value.
- Line charts visualize data trends over time, otherwise known as time series.

EDA with SQL

GitHub Link

SQL Queries

- Display names of the unique launch sites.
- Display 5 records where launch sites began with the string ‘CCA’.
- Display total payload mass carried by boosters launched by NASA (CRS).
- Display the average payload mass carried by booster with version F9 v1.1.
- List the date when the first successful landing outcome for ground pad was achieved.
- List the names of the boosters which had success on drone ship, and have a payload mass greater than 4000 and less than 6000.
- List the total number of successful and failed missions.
- List the names of the booster version which have carried the maximum payload mass.
- List the failed landing outcomes on drone ship, booster versions and launch site names for the months in 2015.
- Rank the count of landing outcomes between 2010-06-04 and 2017-03-20 in descending order.

Build an Interactive Map with Folium

GitHub Link

Launch Site Markers

- Added a marker with a circle, pop-up and text label of NASA Johnson Space Center using latitude/longitude coordinates.
- Added a marker with a circle, pop-up and text label of all launch sites using latitude and longitude coordinates to show proximity to equator and coasts.

Launch Outcome Markers per Launch Site

- Added colored markers for success and failed launches using a marker cluster to visualize which sites have high success rates.

Launch Site Distances

- Added colored lines to indicate distances between launch sites and proximities to a railway, highway, coastline and nearest city.

Build a Dashboard with Plotly Dash

GitHub Link

Drop-down List for Launch Sites

- Drop-down list was added for launch site selection.

Pie Chart Visualizing Successful Launches per Site

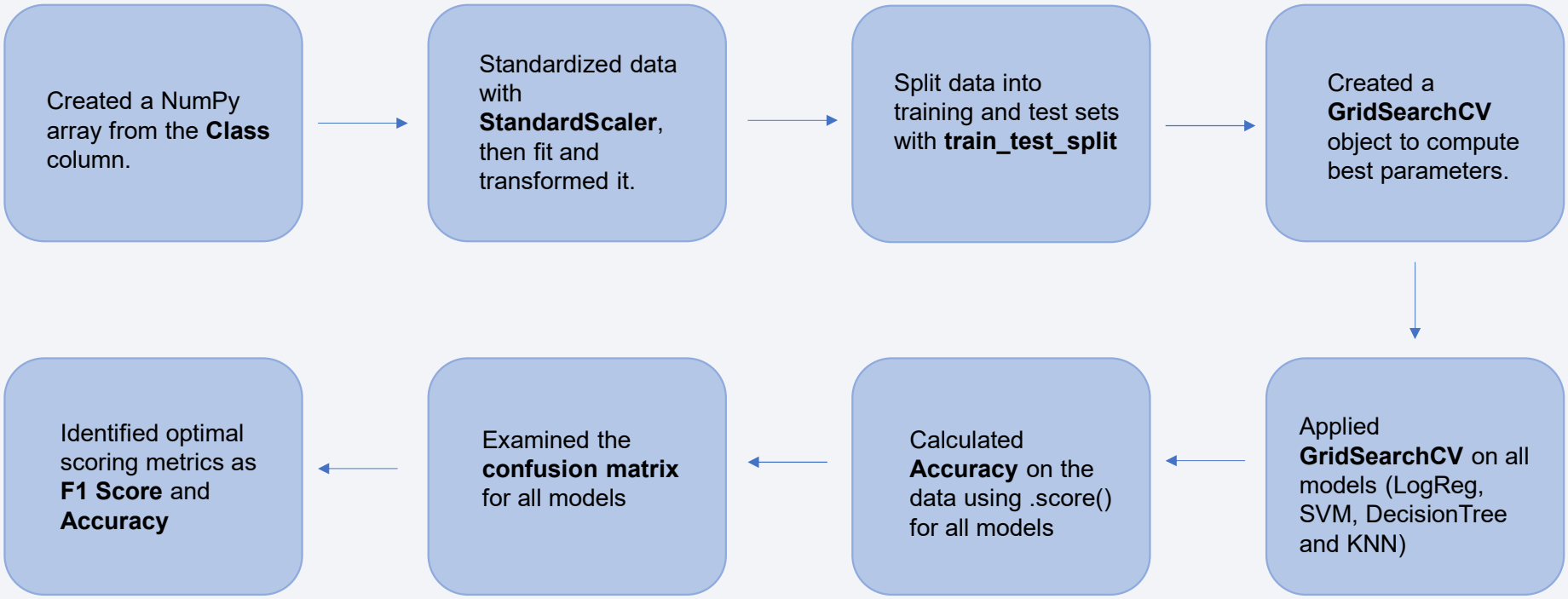
- Pie chart added to show a count of successful launches per site, and a breakdown of successful vs. failed launches when drilling down.

Payload Mass Slider

- Scatter plot added to show the correlation between payload and launch success.

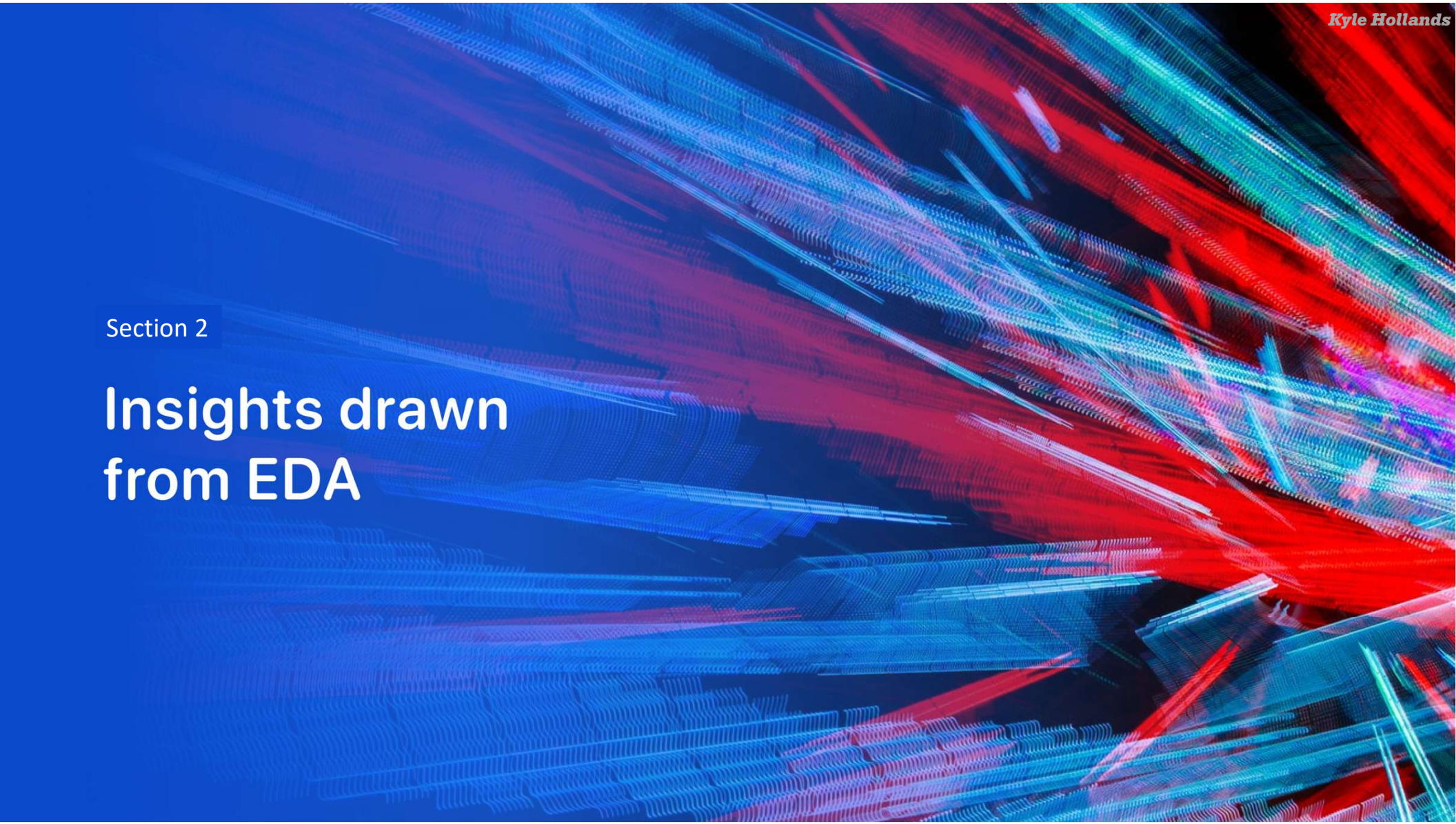
Predictive Analysis (Classification)

GitHub Link



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



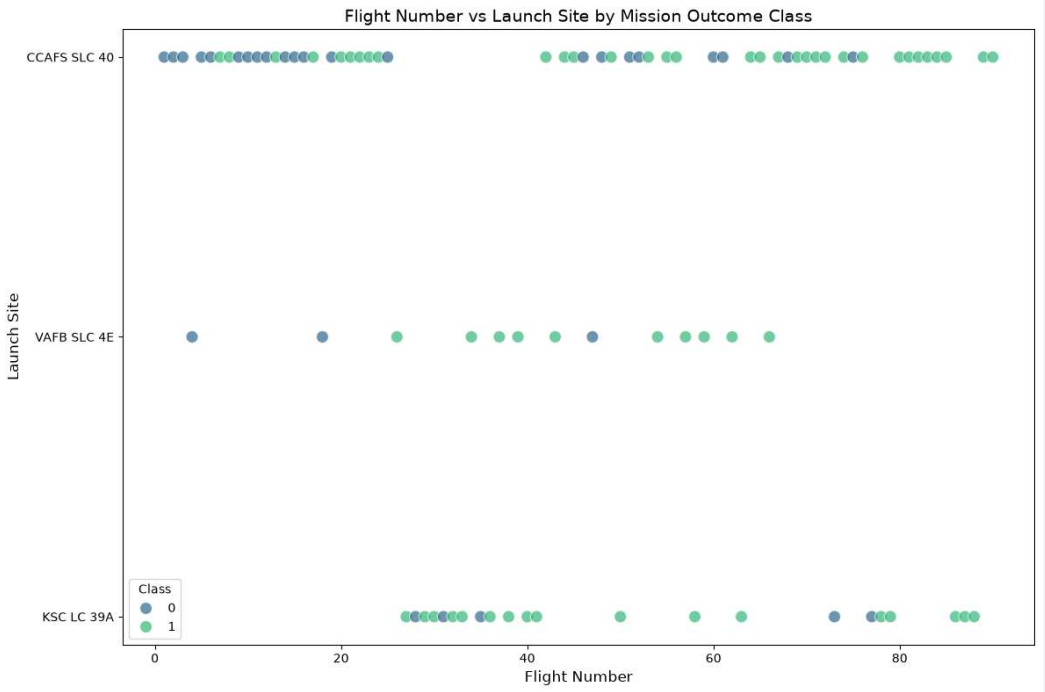
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

GitHub Link

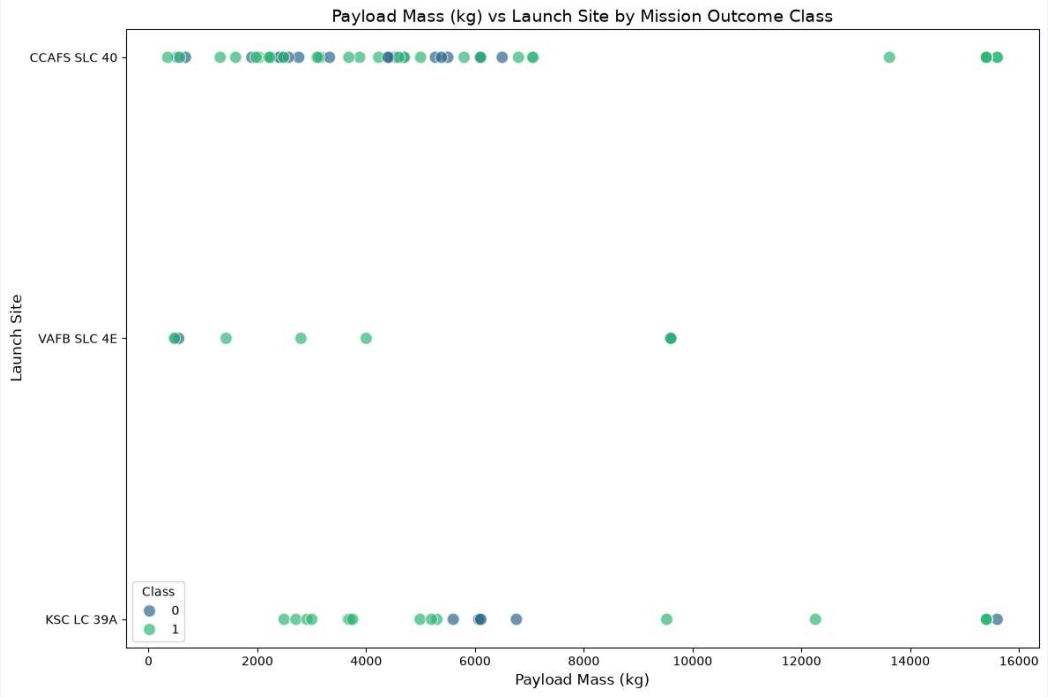
- The earliest flights failed while the latest flights all succeeded.
- The CCAFS SLC 40 launch site has roughly half of all launches.
- VAFB SLC 4E and KSC LC 39A have higher rates of success.
- The assumption can be made that each new launch has a higher rate of success.



Payload vs. Launch Site

GitHub Link

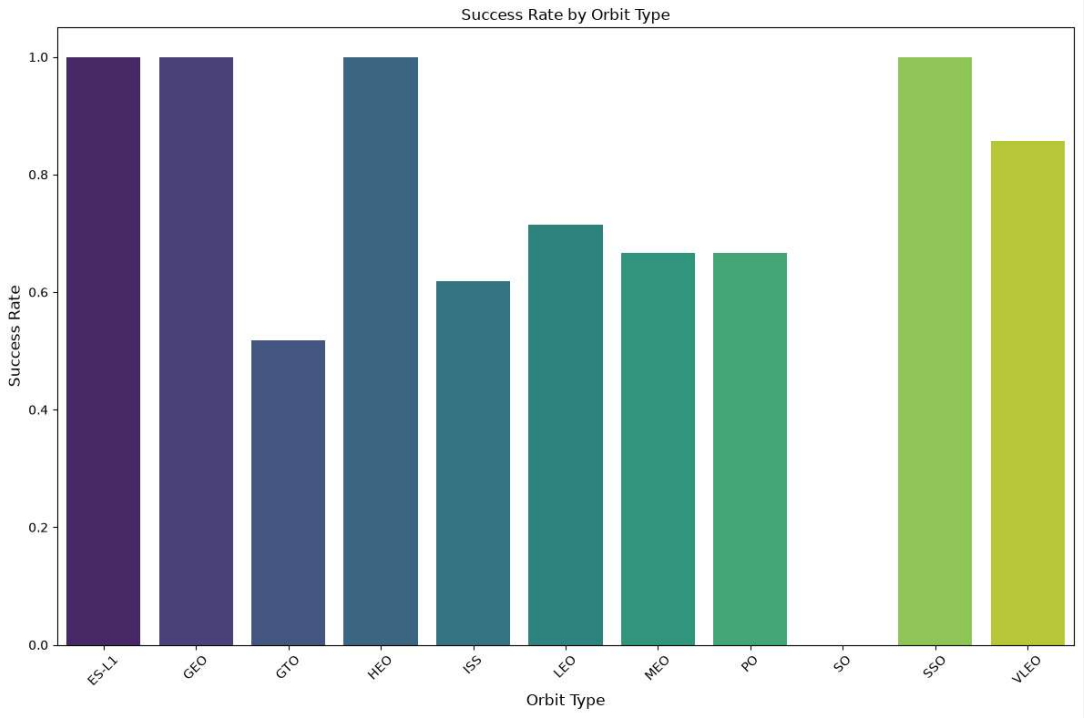
- It appears that as payload mass increases, success rate may increase as well.
- Most launches with payload mass over 7000 kg were successful.
- KSC LC 39A has a 100% success rate when payload mass is less than 5500 kg.



Success Rate vs. Orbit Type

GitHub Link

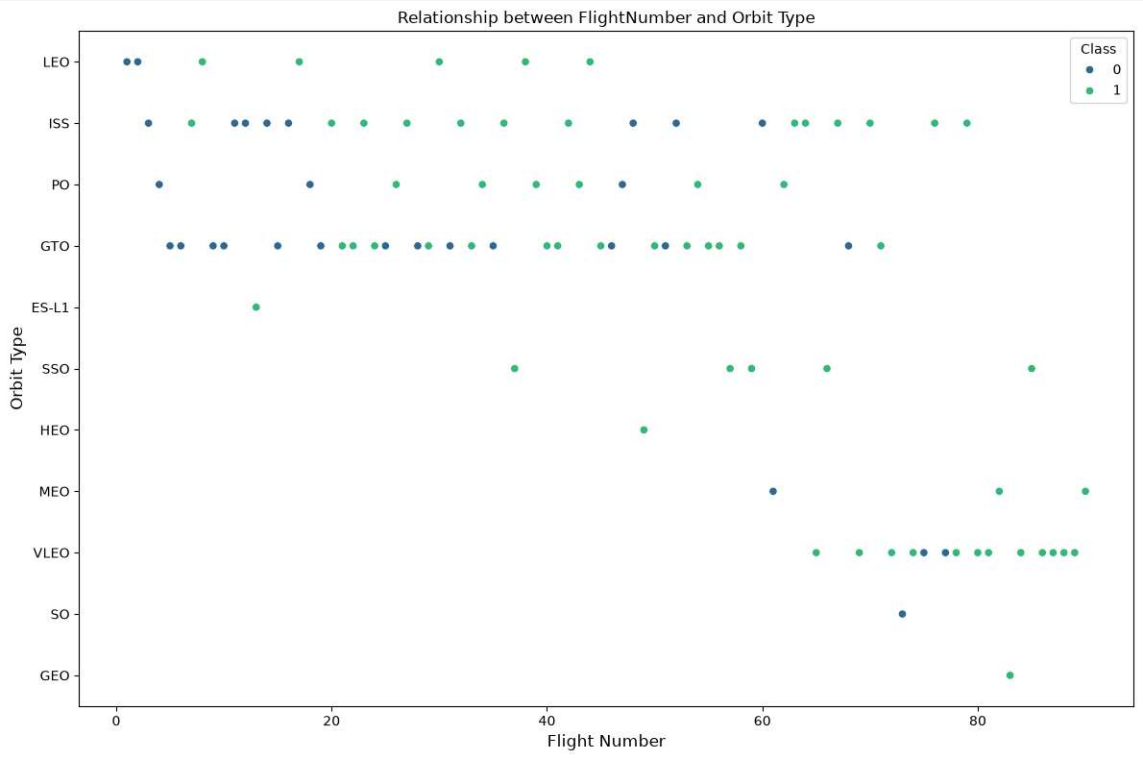
- ES-L1, GEO, HEO and SSO have 100% success rates.
- SO has had a 0% success rate.
- All other orbit types fall between ~50% - ~85% success rates.



Flight Number vs. Orbit Type

GitHub Link

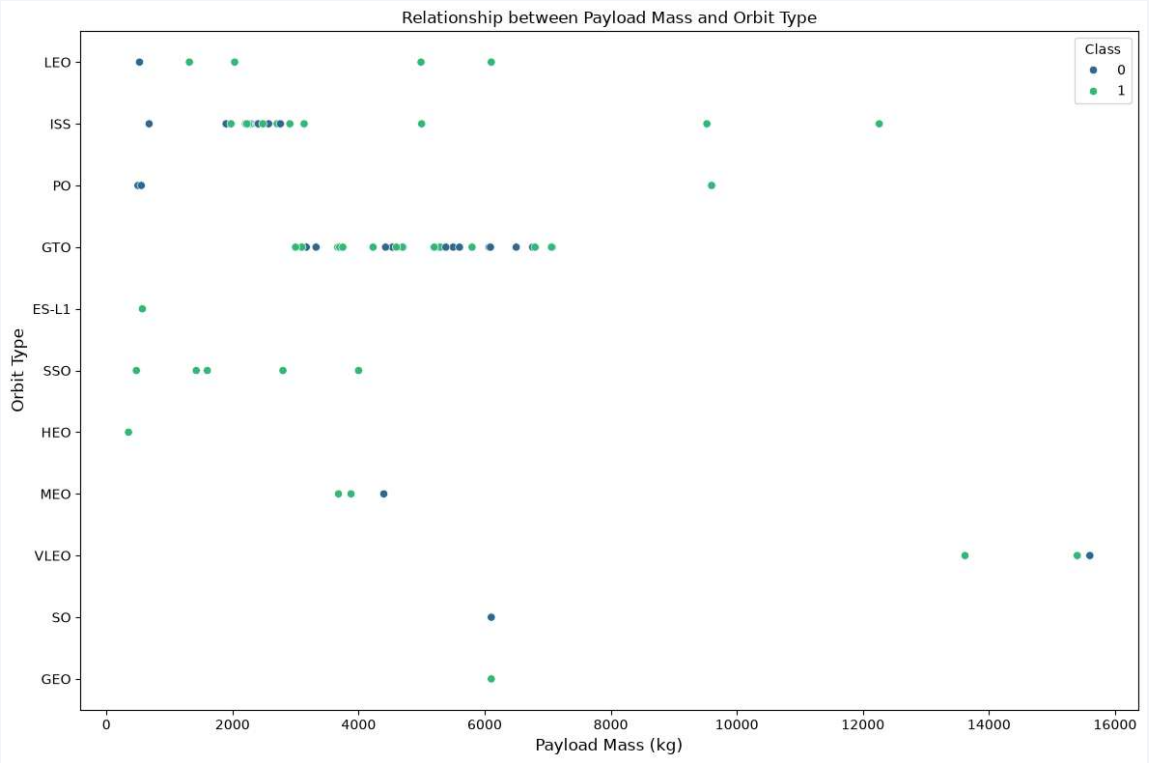
- In the LEO orbit type, success rate appears to be correlated with number of flights. This may also be true for PO. There does not appear to be an obvious correlation between success rate and number of flights for the GTO orbit type.



Payload vs. Orbit Type

GitHub Link

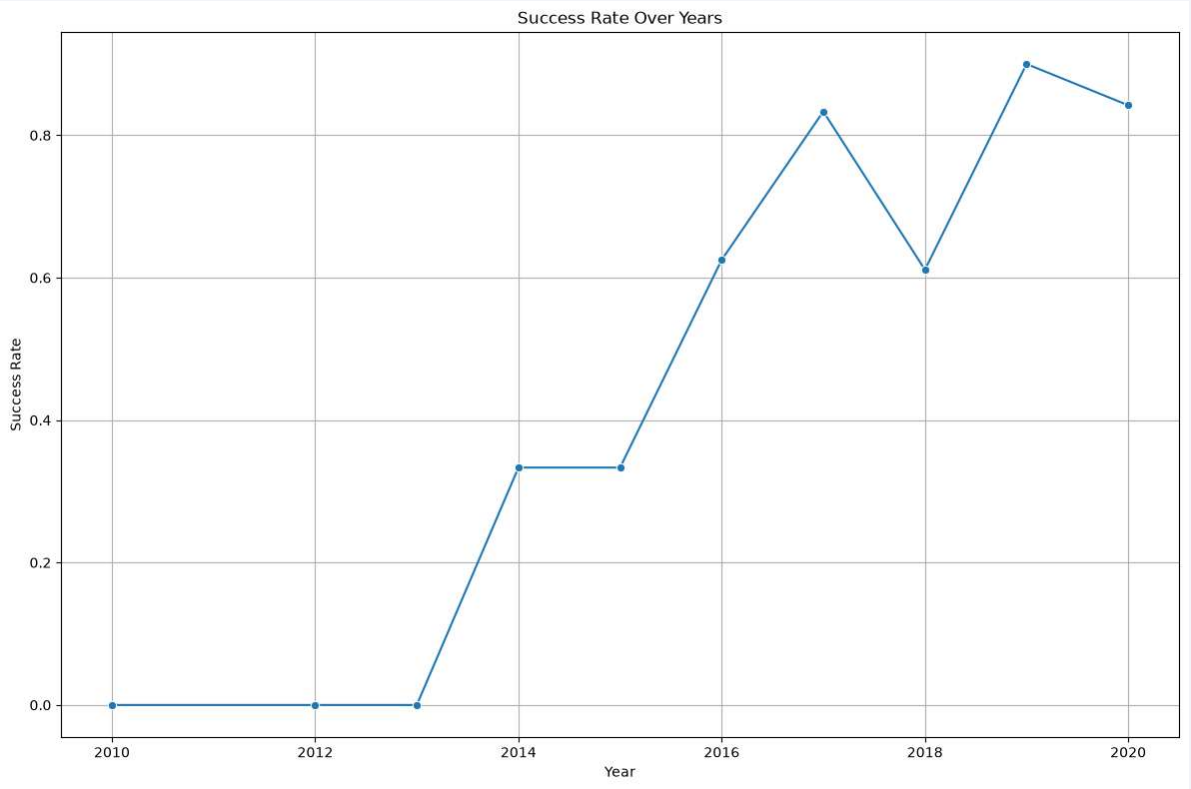
- It appears there may be a correlation between heavier payloads having a negative impact on GTO orbits, while having a positive impact on Polar, LEO and ISS orbits.



Launch Success Yearly Trend

GitHub Link

- It appears the success rate since 2013 climbed continuously until a slight dip in 2018, followed by another increase and another decrease. The overall trend appears to indicate an increasing overall success rate year over year.



All Launch Site Names

GitHub Link

```
# Retrieve distinct launch site names from the SPACEXTABLE dataset
%sql SELECT DISTINCT Launch_Site from SPACEXTABLE
```

[10]

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Utilized SQL to display the names of each unique launch site.

Launch Site Names Begin with 'CCA'

GitHub Link

```
[33]  # Retrieve the first 5 records from the SPACEXTABLE where the Launch_Site begins with 'CCA'
      %sql SELECT * from SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Utilized SQL to display the 5 records where launch sites began with the string ‘CCA’.

Total Payload Mass

GitHub Link

```
# Calculate and display the total payload mass (in kilograms) carried by all boosters launched for NASA's Commercial Resupply Services (CRS) program
%sql SELECT sum(PAYLOAD_MASS__KG_) from SPACEXTABLE WHERE Customer = 'NASA (CRS)'
```

[12]

```
 * sqlite:///my_data1.db
Done.
```

sum(PAYLOAD_MASS__KG_)
45596

- Utilized SQL to display the total payload mass carried by NASA (CSR) launch boosters.

Average Payload Mass by F9 v1.1

GitHub Link

```
[16]  # Calculate average payload mass for F9 v1.1 boosters
      %sql SELECT avg(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'

...   * sqlite:///my_data1.db
      Done.

...   avg(PAYLOAD_MASS__KG_)
      2928.4
```

- Utilized SQL to display the average payload mass carried by F9 v1.1 boosters.

First Successful Ground Landing Date

GitHub Link

```
[19]
    # Find the earliest mission date where both mission outcome and ground pad landing were successful.
    # Select the minimum DATE meeting criteria from SPACEXTABLE.
    %sql SELECT min(DATE) FROM SPACEXTABLE WHERE Mission_Outcome LIKE 'Success%' AND Landing_Outcome = 'Success (ground pad)'

 * sqlite:///my_data1.db
Done.

min(DATE)
2015-12-22
```

- Utilized SQL to list the first successful landing date achieved for ground pad.

Successful Drone Ship Landing with Payload between 4000 and 6000

GitHub Link

```
[21]  # List the booster versions where landing was successful on a drone ship and payload mass is between 4000 and 6000 kg
      %sql SELECT booster_version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;

 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Utilized SQL to list the names of boosters that had successful landings on the drone ship with a payload mass greater than 4000 but less than 6000.

Total Number of Successful and Failure Mission Outcomes

GitHub Link

```
[25]
# Count the number of successful and failed missions from the SPACEXTABLE
%sql SELECT COUNT(CASE WHEN Mission_outcome LIKE 'Success%' THEN 1 END) AS success_count,COUNT(CASE WHEN Mission_outcome LIKE 'Failure%' THEN 1 END) AS failure_count FROM SPACEXTABLE;

 * sqlite:///my_data1.db
Done.
```

success_count	failure_count
100	1

- Utilized SQL to list the total number of successful and failed mission outcomes.

Boosters Carried Maximum Payload

GitHub Link

```
[30]  # Retrieve booster version with the maximum payload mass
      %sql SELECT booster_version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT max(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);

...   * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- Utilized SQL to list the names of all booster versions that have carried the maximum payload mass.

2015 Launch Records

GitHub Link

```
# Retrieve details of failed landings in 2015
%sql SELECT substr(Date, 6,2) AS month, landing_outcome, booster_version, launch_site FROM SPACEXTABLE WHERE substr(Date,0,5)='2015' AND landing_outcome LIKE 'Failure%';
```

```
 * sqlite:///my_data1.db
Done.
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Utilized SQL to display failed landing outcomes for drone ship, their corresponding booster versions and launch site for the year 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

GitHub Link

```
[32]
    # Execute query to count landing outcomes between specific dates
    %sql SELECT landing_outcome, count(*) AS number FROM SPACEXTABLE WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY landing_outcome ORDER BY number DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	number
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Utilized SQL to display the various landing outcomes, ranked by descending order between the dates of 2010-06-04 and 2017-03-20.



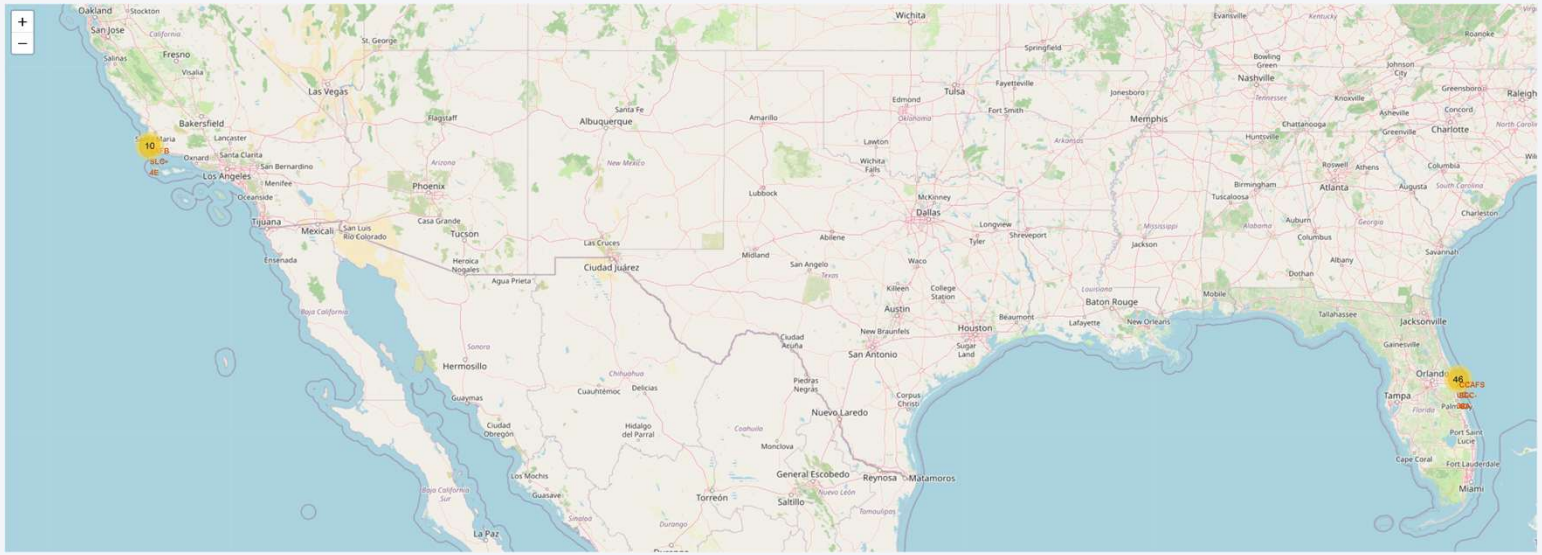
Section 3

Launch Sites Proximities Analysis

All Launch Sites

GitHub Link

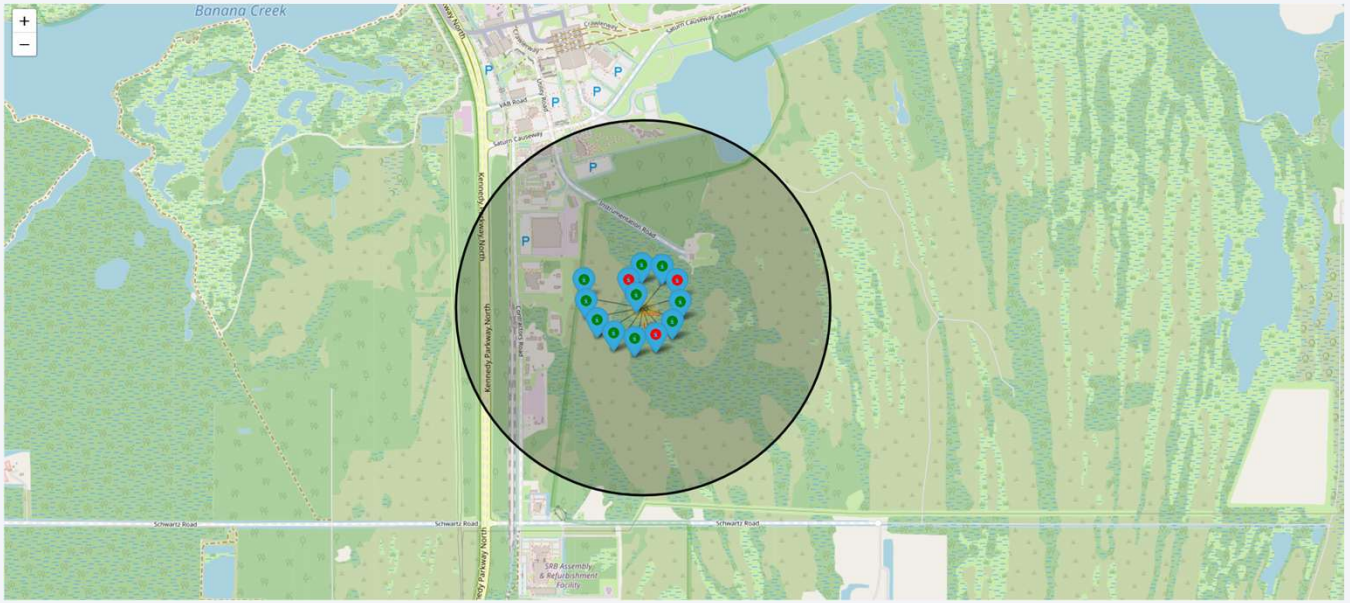
- Most launch sites are in proximity to the equator, which is a strategic decision to optimize the efficiency of launches due to the speed of Earths rotation at the equator being greater than anywhere else on Earth. Additionally, all launch sites are near costs, which minimizes any potential damage to populated areas.



Colored Launch Markers

GitHub Link

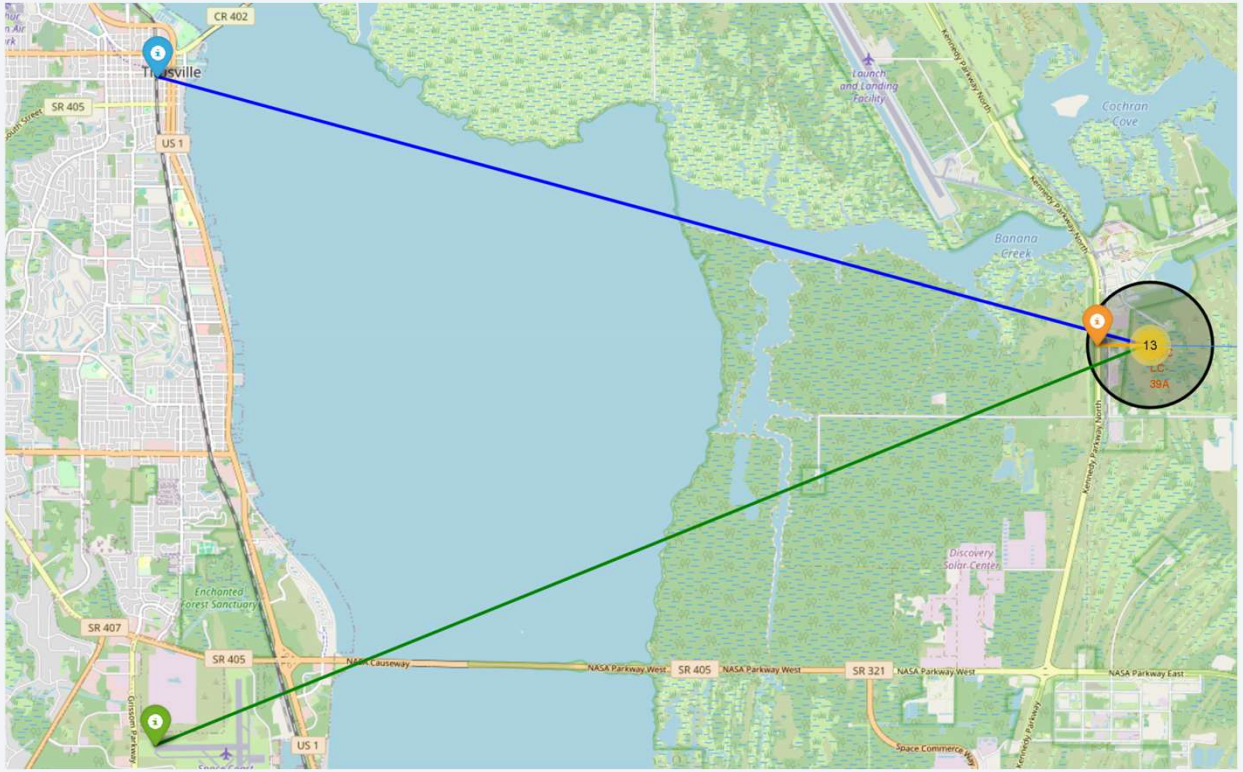
- The colored markers allow us to easily identify the number of successful and failed launches at each launch site. Launch site KSC LC-39A has a very high success rate.

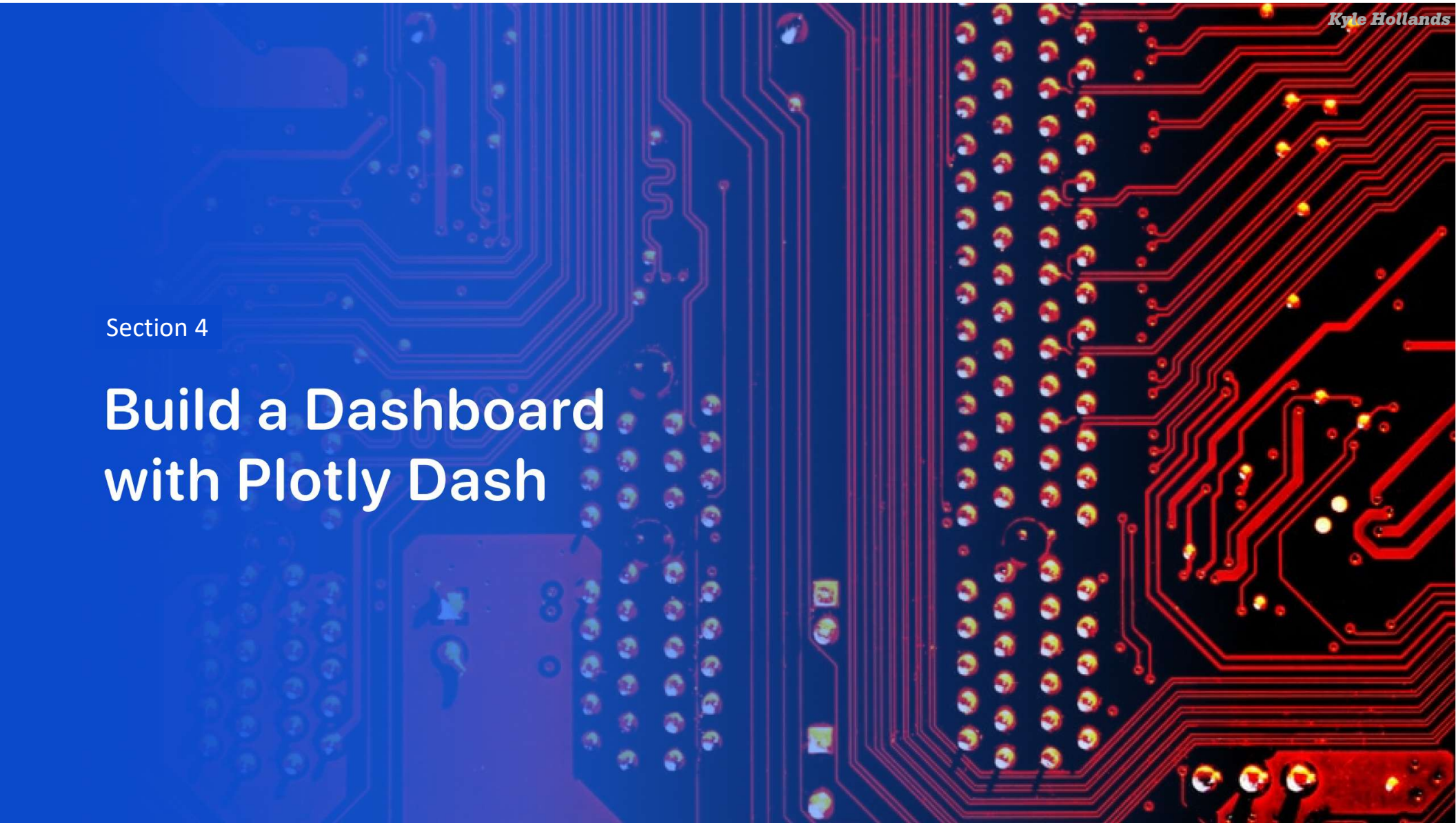


Launch Site KSC LC-39A to Proximities

GitHub Link

- The following map displays the relative distance from the chosen launch site (KSC LC-39A) and nearby major proximities, such as the nearest railway, highway and city.



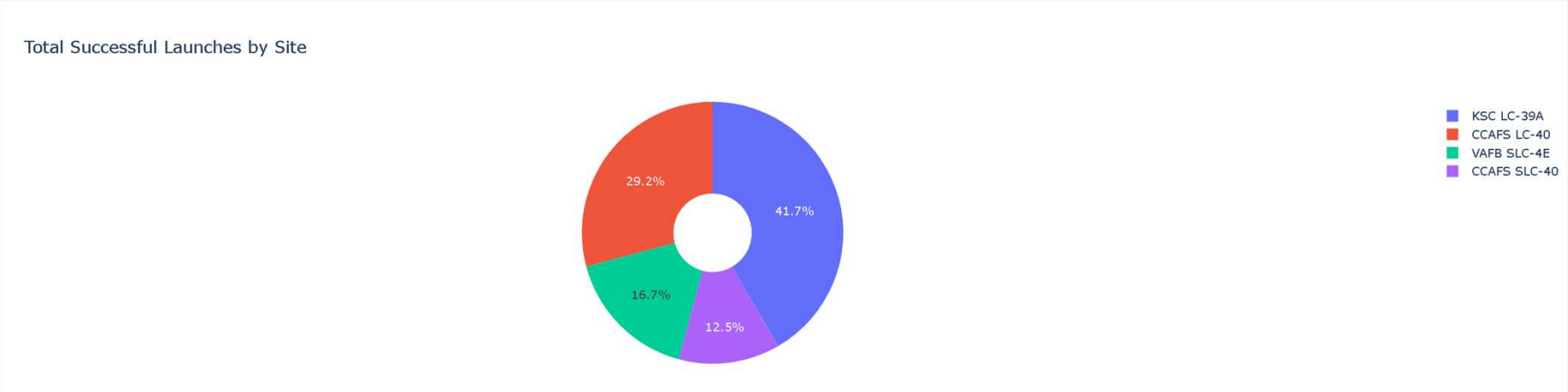


Section 4

Build a Dashboard with Plotly Dash

Count of all Successful Launches for all Sites

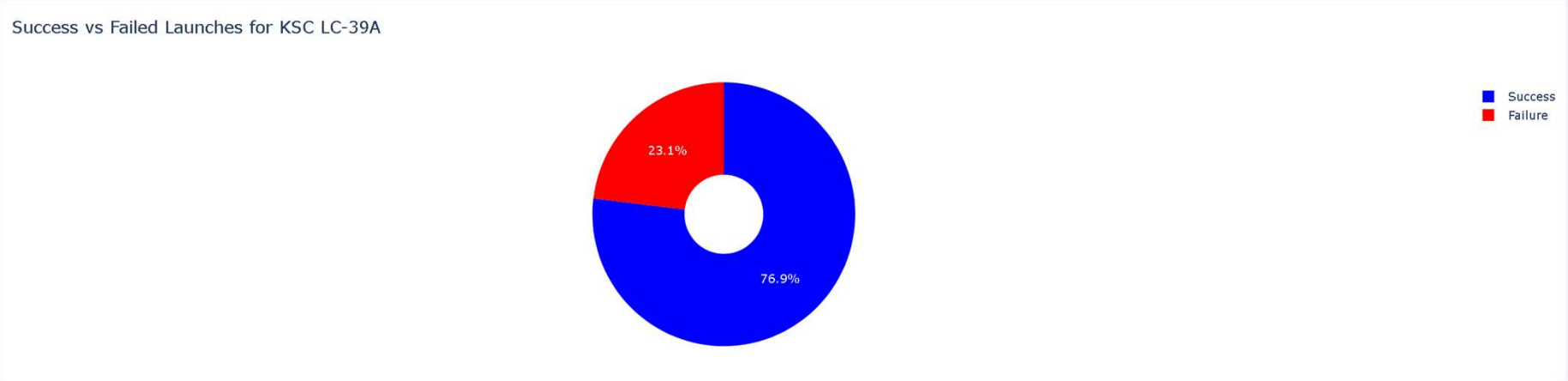
GitHub Link



- The chart indicates that of all launch sites, KSC LC-39A has the most successful launches, which is expected based on prior analysis.

Highest Launch Site Success Ratio

GitHub Link



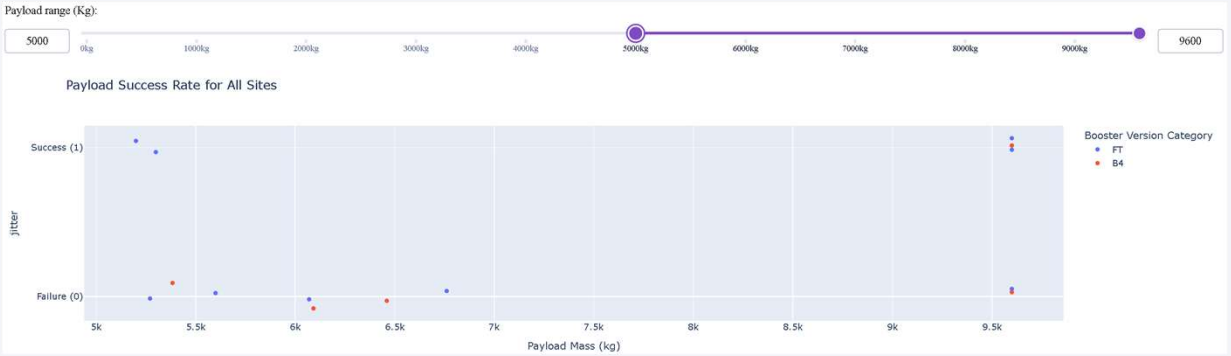
- This drilled-down donut chart visualizes the proportion of successful vs failed launches at the KSC LC-39A launch site.

Payload Mass vs. Launch Outcome (All Sites)

GitHub Link

- The plots appear to indicate that payloads between 2000kg and 5000kg achieve the highest success rate.







Section 5

Predictive Analysis (Classification)

Classification Accuracy

GitHub Link

- The accuracy scores on the test set all appeared to be the same (~0.83), potentially due to the very small sample size being tested.
- Based on the entirety of the model, it appears the Support Vector Machines model performs best with a score of ~87%.

Find the method performs best:

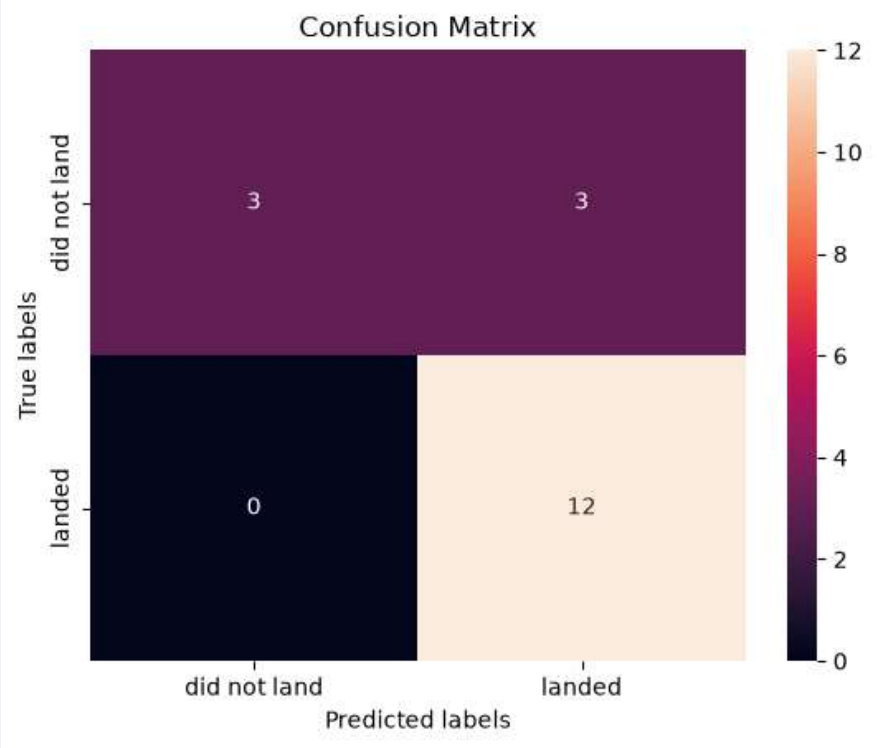
```
[126] # Compare results - this line runs after each model
      results_df = pd.DataFrame(model_results).sort_values('best_score', ascending=False)
      results_df
```

	model	best_params	best_score
1	Support Vector Machines	{'C': 1.0, 'class_weight': None, 'coef0': 1.77...	0.875000
0	Logistic Regression	{'C': 0.1, 'class_weight': {0: 1, 1: 2}, 'fit_...	0.862500
2	Decision Tree	{'criterion': 'gini', 'max_depth': 6, 'max_fea...	0.846429
3	K-Nearest Neighbours	{'algorithm': 'auto', 'n_neighbors': 7, 'p': 1...	0.846429

Confusion Matrix

GitHub Link

- The Confusion Matrix for the Support Vector Machines model indicates 3 false positives, or in other words, incorrectly classified 3 observations as landed, when it did not



Conclusions

- The Support Vector Machines performed best on the dataset.
- The data appears to indicate that larger payload masses may result in more negative launch outcomes.
- Most of the launch sites are close to the equator, and all sites are close to a coast.
- The success rate for launches appears to increase year over year, and this trend is likely to continue.
- KSC lC-39A has the highest success rate of all launch sites.
- ES-L1, GEO, HEO and SSO have 100% success rates.

Thank you!

